FINITE DIMENSIONALITY OF TANGENT AND INFINITESIMAL AUTOMORPHISM SPACES OF THE COHERENT STACK

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-coherent (canonical tag 08KB). cited at statement line 1431. Standing situation.
- Section artin-section-tangent-spaces (canonical tag 07WY). cited at statement line 1438. Definitions of tangent space and Inf.
- Lemma lemma-coherent-RS-star (canonical tag 08LQ). cited at proof lines 1443-1444. (RS*) of the coherent stack.
- Section artin-section-inf (canonical tag 07Y6). cited at proof line 1445. Definition of (RS*).
- Lemma artin-lemma-deformation-category (canonical tag 07WU). cited at proof lines 1448-1449. Fibres are deformation categories.
- Remark remark-coherent-base-change (canonical tag 08LP). cited at proof line 1456. X_0's stack as 2-fibre product.
- Lemma artin-lemma-fibre-product-tangent-spaces (canonical tag 07X2). cited at proof lines 1459-1460. Tangent/Inf of 2-fibre products.
- Lemma artin-lemma-finite-dimension (canonical tag 07X1). cited at proof lines 1463-1464. Finite dimensionality for B and Spec(k).
- Lemma lemma-coherent-over-first-order-thickening (canonical tag 08W7). cited at proof lines 1477-1478. Tangent space = lifts of G_0.
- Lemma defos-lemma-flat-ringed-topoi (canonical tag 08VW). cited at proof lines 1481 and 1489. Ext^1 computes T F; Inf = Ext^0.
- Lemma spaces-perfect-lemma-ext-finite (canonical tag 0D0T). cited at proof lines 1499-1500. Finiteness of the Ext groups.

Source reference. Original definition 08KB: sources/stacks-chow/tags.txt, source lines 10879–10879. Cited at statement line 1431: the lemma is phrased in this situation; load-bearing.

Source reference. Original section 07WY: sources/stacks-chow/tags.txt, source lines 10058–10058. Cited at statement line 1438: the objects whose finite dimensionality is claimed; load-bearing.

Date: stacks-project:quot.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08LQ: sources/stacks-chow/tags.txt, source lines 10928–10928. Cited at proof lines 1443-1444: X satisfies (RS*), hence (RS); load-bearing.

Source reference. Original section 07Y6: sources/stacks-chow/tags.txt, source lines 10101–10101. Cited at proof line 1445: definition of (RS*); load-bearing.

Source reference. Original lemma 07WU: sources/stacks-chow/tags.txt, source lines 10054–10054. Cited at proof lines 1448-1449: the fibres are deformation categories; load-bearing.

Source reference. Original remark 08LP: sources/stacks-chow/tags.txt, source lines 10927–10927. Cited at proof line 1456: X_0's stack is the 2-fibre product with Spec(k) over B; load-bearing for the reduction.

Source reference. Original lemma 07X2: sources/stacks-chow/tags.txt, source lines 10062–10062. Cited at proof lines 1459-1460: reduces to the three factors; load-bearing.

Source reference. Original lemma 07X1: sources/stacks-chow/tags.txt, source lines 10061–10061. Cited at proof lines 1463-1464: B and Spec(k) have finite dimensional tangent spaces; load-bearing.

Source reference. Original lemma 08W7: sources/stacks-chow/tags.txt, source lines 11260–11260. Cited at proof lines 1477-1478: the tangent space is the set of lifts of G_0 to a flat module on X_epsilon; load-bearing.

Source reference. Original lemma 08VW: sources/stacks-chow/tags.txt, source lines 11249–11249. Cited at proof lines 1481 and 1489 (twice): T F = Ext^1 and Inf = Ext^0; load-bearing twice.

Source reference. Original lemma 0D0T: sources/stacks-chow/tags.txt, source lines 16323–16323. Cited at proof lines 1499-1500: G_0 coherent on the Noetherian X_0 gives finite dimensional Exts; load-bearing.

Lemma 0.2. *In Situation 08KB assume that S is a locally Noetherian scheme and $B \to S$ is locally of finite presentation. Let k be a finite type field over S and let $x_0 = (\mathrm{Spec}(k), g_0, \mathcal{G}_0)$ be an object of $\mathcal{X} = \mathcal{C}oh_{X/B}$ over k. Then the spaces $T\mathcal{F}_{\mathcal{X},k,x_0}$ and $Inf(\mathcal{F}_{\mathcal{X},k,x_0})$ (Artin's Axioms, Section 07WY) are finite dimensional.*

Proof. Observe that by Lemma 08LQ our stack in groupoids $\mathcal{X}$ satisfies property (RS*) defined in Artin's Axioms, Section 07Y6. In particular $\mathcal{X}$ satisfies (RS). Hence all associated predeformation categories are deformation categories (Artin's Axioms, Lemma 07WU) and the statement makes sense.

In this paragraph we show that we can reduce to the case $B = \mathrm{Spec}(k)$. Set $X_0 = \mathrm{Spec}(k) \times_{g_0,B} X$ and denote $\mathcal{X}_0 = \mathcal{C}oh_{X_0/k}$. In Remark 08LP we have seen that $\mathcal{X}_0$ is the 2-fibre product of $\mathcal{X}$ with $\mathrm{Spec}(k)$ over B as categories fibred in groupoids over $(Sch/S)_{fppf}$. Thus by Artin's Axioms, Lemma 07X2 we reduce to proving that B, $\mathrm{Spec}(k)$, and $\mathcal{X}_0$ have finite dimensional tangent spaces and infinitesimal automorphism spaces. The tangent space of B and $\mathrm{Spec}(k)$ are finite dimensional by Artin's Axioms, Lemma 07X1 and of course these have vanishing Inf. Thus it suffices to deal with $\mathcal{X}_0$.

Let $k[\epsilon]$ be the dual numbers over k. Let $\mathrm{Spec}(k[\epsilon]) \to B$ be the composition of $g_0 : \mathrm{Spec}(k) \to B$ and the morphism $\mathrm{Spec}(k[\epsilon]) \to \mathrm{Spec}(k)$ coming from the inclusion

$k \to k[\epsilon]$. Set $X_0 = \mathrm{Spec}(k) \times_B X$ and $X_\epsilon = \mathrm{Spec}(k[\epsilon]) \times_B X$. Observe that X_ϵ is a first order thickening of X_0 flat over the first order thickening $\mathrm{Spec}(k) \to \mathrm{Spec}(k[\epsilon])$. Unwinding the definitions and using Lemma 08W7 we see that $T\mathcal{F}_{\mathcal{X}_0,k,x_0}$ is the set of lifts of $\mathcal{G}_0$ to a flat module on X_ϵ. By Deformation Theory, Lemma 08VW we conclude that

$$T\mathcal{F}_{\mathcal{X}_0,k,x_0} = \mathrm{Ext}^1_{\mathcal{O}_{X_0}}(\mathcal{G}_0, \mathcal{G}_0)$$

Here we have used the identification $\epsilon k[\epsilon] \cong k$ of $k[\epsilon]$-modules. Using Deformation Theory, Lemma 08VW once more we see that

$$\mathrm{Inf}(\mathcal{F}_{\mathcal{X},k,x_0}) = \mathrm{Ext}^0_{\mathcal{O}_{X_0}}(\mathcal{G}_0, \mathcal{G}_0)$$

These spaces are finite dimensional over k as $\mathcal{G}_0$ has support proper over $\mathrm{Spec}(k)$. Namely, X_0 is of finite presentation over $\mathrm{Spec}(k)$, hence Noetherian. Since $\mathcal{G}_0$ is of finite presentation it is a coherent $\mathcal{O}_{X_0}$-module. Thus we may apply Derived Categories of Spaces, Lemma 0D0T to conclude the desired finiteness. □